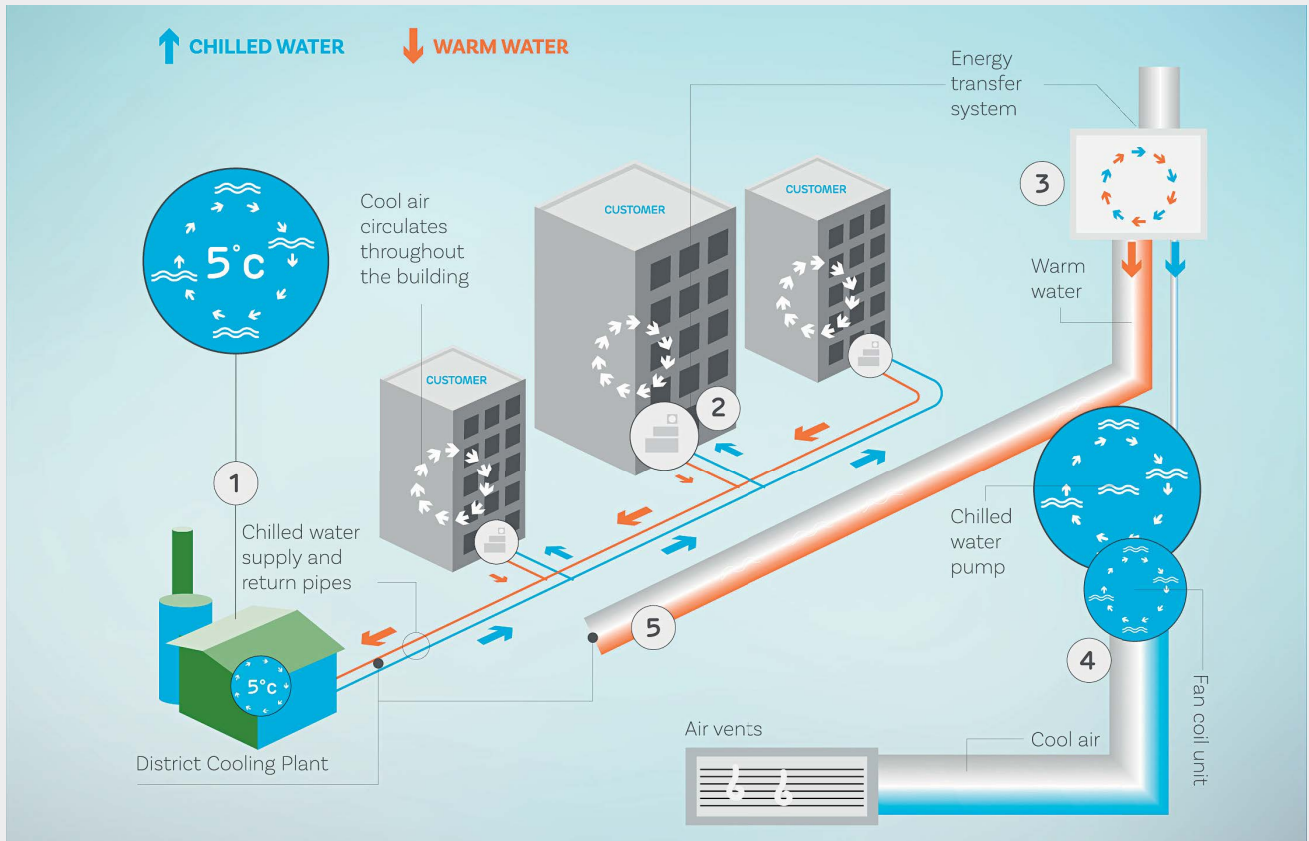




with grid power, making DCS solutions more affordable and sustainable.

A major benefit of large-scale district cooling systems (DCS) comes from the diversity factor of peak consumption when combined with mixed-load developments. This helps flatten the demand curve, thereby enhancing the loading capacity of transmission and distribution lines.

Adani's Unique DCS Solution

For the Adani Group, DCS aligns with its focus on creating new infrastructure that is sustainable, more efficient and aligned to India's net zero emissions vision. For AESL, DCS will not only enhance energy efficiency and improve grid stability, but also ensure optimised cost and reduced carbon emissions. DCS can shave off peak demand, optimise resource use, and reduce the overall power demand on distribution networks. By integrating with distribution services, it can be seen as a win-win value-added service.

AESL's DCS solution is offered as "cooling-as-a-service", where AESL bears the entire capital expenditure, with the user only paying based on usage. This makes

AESL's proposition highly attractive for the developer as it frees up significant portion of their HVAC (heating, cooling, ventilation, and air conditioning) investments and costly commercial space for sale.

AESL is currently implementing a couple of projects and plans to scale up the business significantly in the coming years.

Indian Market Opportunity

As per India Cooling Action Plan 2019 and District Cooling Guidelines 2023, the cooling requirement of India is estimated to grow 4x from 130 Million Ton of Refrigeration (MTR) in 2022-23 to 720 MTR by 2037-38. Three fourths of this requirement would be from space cooling needs of housing and realty infrastructure, requiring energy consumption of around 476 TWh, translating to an additional 120 GW of power generation and corresponding increase in transmission and distribution network.

Studies by the IEA, World Economic Forum, and ICAP show that India's per capita cooling energy consumption in 2023 was about 147 kWh, compared to

the global average of 1,539 kWh. Also, the penetration in AC ownership in India is around 8% as compared to 60 to 90% in other developing countries, even though the person Cooling Degree Days (CDD) of India is highest at over 4,200 billion due to its large population and year-round high ambient temperature across large swathes of the country.

With significant development happening in realty and urban housing development projects, the addressable market for DCS in India is expected to grow by almost 20% CAGR from meagre 9 MTR in 2023 to around 132 MTR by 2037-38 with market share increasing 3x from 7% to 21%, with the share of single room ACs and VRF (Variable Refrigerant Flow) systems dropping from the current 90% to 78% over the same period.

The growth in urbanisation and industrialisation will increase the demand for efficient and sustainable cooling solutions, driving the expansion of DCS. As technology advances and market awareness increases, DCS is going to play a vital role in creating sustainable and resilient urban environments.